

Processando config 19, seed 1, teste: Alphabit
xx
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
xx

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = $2^L = 4$
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	2
Value of the statistic	:	1.03
p-value of test	:	0.60

CPU time used : 00:00:00.01

Generator state:

16777216 bits have been read.

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Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = $2^L = 16$
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 9.26
p-value of test : 0.32

CPU time used : 00:00:00.03

Generator state:
33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

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smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 130.30
p-value of test : 0.43

CPU time used : 00:00:00.01

Generator state:
50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

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smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 33046.63
p-value of test : 0.14

CPU time used : 00:00:00.01

Generator state:

67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

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sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 207.47
p-value of test : 0.06

CPU time used : 00:00:00.01

Generator state:

83886080 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 309.55
p-value of test : 0.42

CPU time used : 00:00:00.01

Generator state:

100663296 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 0.82
p-value of test : 0.21

CPU time used : 00:00:00.01

Generator state:

117440512 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom : 32
ChiSquare statistic : 34.41
p-value of test : 0.35

Test on the values of the Statistic M

Number of degrees of freedom : 31
ChiSquare statistic : 22.71
p-value of test : 0.86

Test on the values of the Statistic J

Number of degrees of freedom : 32
ChiSquare statistic : 31.35
p-value of test : 0.50

Test on the values of the Statistic R

Number of degrees of freedom : 25
ChiSquare statistic : 18.78
p-value of test : 0.81

Test on the values of the Statistic C

Number of degrees of freedom : 15
ChiSquare statistic : 32.37
p-value of test : 5.7e-3

CPU time used : 00:00:00.07

Generator state:
134217728 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

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swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom : 62

ChiSquare statistic : 64.91
p-value of test : 0.38

Test on the values of the Statistic M

Number of degrees of freedom : 62
ChiSquare statistic : 54.13
p-value of test : 0.75

Test on the values of the Statistic J

Number of degrees of freedom : 160
ChiSquare statistic : 158.14
p-value of test : 0.53

Test on the values of the Statistic R

Number of degrees of freedom : 57
ChiSquare statistic : 56.98
p-value of test : 0.48

Test on the values of the Statistic C

Number of degrees of freedom : 32
ChiSquare statistic : 25.53
p-value of test : 0.78

CPU time used : 00:00:00.05

Generator state:
150994688 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.21

All tests were passed